A SYNTOMIC ALGEBRA MAPS SMOOTHLY, WITH A RETRACTION, TO A RELATIVE GLOBAL COMPLETE INTERSECTION

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Editorial note. Input lemma-improve-presentation is a published entry (tag 07CE) used as a standing input by its display number; not proven on the author's behalf.

Allowed standard prerequisites (editorial).

- Presentation improvement with retraction. lemma-improve-presentation (same chapter, PUBLISHED 07CE). Produces A -> C with retraction, C smooth over A, and freeness of J/J^2 at lci points.
- Huber's presentation lemma. Algebra, lemma huber (cross-chapter, named). Chooses generators f_1, ..., f_c of the presentation ideal mapping to a basis of J/J^2.
- Fibre dimension of a local complete intersection. Algebra, lemma lci (cross-chapter, named). Fibres of C over R are lci, so they have dimension n - c.

Source reference. Original reference 07CF: focused-seeds/geometry/S9-smo-syntomic-complete-intersection/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00SC: focused-seeds/geometry/S9-smo-syntomic-complete-intersection/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07CE: sources/stacks-smoothing/pinned-original-source.tex, source lines 559–559. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $R \to A$ be a syntomic ring map. Then there exists a smooth R-algebra map $A \to C$ with a retraction such that C is a global relative complete intersection over R, i.e.,*

$$C \cong R[x_1, \ldots, x_n]/(f_1, \ldots, f_c)$$

flat over R and all fibres of dimension $n - c$.

Date: stacks-project:smoothing.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Apply Lemma 07CE to get $A \to C$. By Algebra, Lemma 07CF we can write $C = R[x_1, \ldots, x_n]/(f_1, \ldots, f_c)$ with f_i mapping to a basis of J/J^2. The ring map $R \to C$ is syntomic (hence flat) as it is a composition of a syntomic and a smooth ring map. The dimension of the fibres is $n - c$ by Algebra, Lemma 00SC (the fibres are local complete intersections, so the lemma applies). $\square$